Notes on the Harmonic Oscillator in Phase Space

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Abstract

These notes collect the standard results on the quantum harmonic oscillator and compare them with the classical phase-space picture [2, 1].

1 Introduction

The harmonic oscillator is the first system every course solves twice: once with differential equations and once with ladder operators.

2 Ladder operators

Define the annihilation operator

$$a = \sqrt{\frac{m\omega}{2\hbar}}\left(x + \frac{ip}{m\omega}\right), \tag{1}$$

so that the Hamiltonian becomes

$$H = \hbar\omega\left(a^\dagger a + \tfrac{1}{2}\right). \tag{2}$$

The commutator $[a, a^\dagger] = 1$ follows from $[x, p] = i\hbar$.

3 Phase-space picture

Figure 1 shows the classical trajectory as an ellipse whose area grows with the energy, matching the Bohr–Sommerfeld rule [1].

4 Conclusion

Equations (1)–(**??**) are all one needs for most applications.

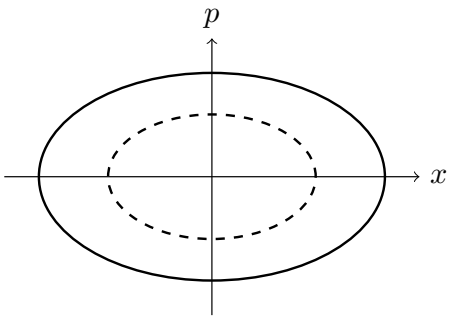


Figure 1: Classical phase-space orbit of the oscillator.

Level	Energy	Parity
$n = 0$	$\frac{1}{2}\hbar\omega$	even
$n = 1$	$\frac{3}{2}\hbar\omega$	odd

Table 1: The two lowest levels.

References

[1] David J. Griffiths and Darrell F. Schroeter. *Introduction to Quantum Mechanics*. Cambridge University Press, 3 edition, 2018.

[2] J. J. Sakurai and Jim Napolitano. *Modern Quantum Mechanics*. Cambridge University Press, 2 edition, 2017.